



Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 421: Software Engineering

Batch: 22nd

Date: 27/02/2026

Final Examination

Semester: Fall 2025

Time: 2 Hours

Total Marks : 40

Answer any five questions:

5 x 8 = 40

1. (a) Write the principles that guide a Software Process. 4
(b) Develop a complete use case diagram for "Making a withdrawal from an ATM". 4
2. (a) Explain all seven tasks for requirements engineering. 4
(b) Illustrate a software requirements specification template. 4
3. (a) Explain domain analysis in requirement modeling. 4
(b) Briefly explain the requirement modeling approaches. 4
4. (a) How do we assess the quality of a software design? 4
(b) What are the main uses of the requirement models? 4
5. (a) Draw the way of translation of requirements model into the design model. 4
(b) Describe software quality guidelines and attributes. 4
6. (a) Explain the golden rules for user interface design. 4
(b) What is interface analysis? Explain. 4
7. (a) Explain WebApp design quality and quality requirements tree. 4
(b) Illustrate a design pyramid for WebApp. 4



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering

CSE423: Simulation and Modeling

Batch: 22nd

Date: 01-03-2026

Final Examination

Semester: Fall 2025

Time: 2 Hours

Total Marks: 40

Answer any **five** questions of the followings.

5 × 8 = 40

1. (a) Explain the types of simulations with respect to output analysis with example. 4
(b) Explain Inverse Transform Method for generating random variates with example. 4
2. (a) Write the steps involved in the development of a useful model of input data. 4
(b) What is input modeling in simulation? Why is input modelling important in simulation? 4
3. (a) When is a stochastic process called a Continuous-Time Markov Chain? 3
(b) What is meant by Confidence Interval Overlap in simulation output comparison? 5
Explain how confidence interval overlap is used to compare two systems.
4. (a) List the issues that are considered in an accreditation decision. 3
(b) Draw the diagram for event scheduling scheme in computer simulation. 5
5. (a) What is Monte Carlo Simulation? What kind of input data is required for performing Monte Carlo simulations? 3
(b) Use Monte Carlo Simulation to approximate the value of $\int_3^5 x^2 dx$ 5
R1 → 0.21, 0.11, 0.71, 0.65, 0.41, 0.35, 0.17
R2 → 0.96, 0.82, 0.56, 0.68, 0.58, 0.40, 0.82
6. Generate a sequence of 5 two-digit random numbers by employing the following methods:
 - (i) Mixed Congruential Method 2
 - (ii) Multiplicative Congruential Method 2
 - (iii) Additive Congruential Method- 2
The recursive equation for the mixed congruential method is-
 $R_{i+1} = (21R_i + 53) \text{ (module 100) and seed } R_0 = 46$
 - (iv) Midsquare Method (using same seed) 2



HAMMARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 469 : Digital Image Processing

Batch: 22nd

Date: 06/03/2026

Final Examination

Semester: Fall 2025

Time: 2 Hours

Total Marks: 40

Answer any five questions.

5 × 8 = 40

1. (a) Define Quantitative and Structural Patterns. 2
(b) Explain how Deep Convolutional Neural Networks can learn pattern features directly from training data images. 6
2. (a) Briefly describe different types of Data Redundancy in Images. 3
(b) Determine the average length of the code from the following information after source reduction using the Huffman Coding algorithm. 5

Original Source	
Symbol	Probability
A1	0.1
A2	0.4
A3	0.06
A4	0.1
A5	0.04
A6	0.3

3. (a) Differentiate between lossless and lossy compression system. 2
(b) A three-symbol source has the following probabilities: 6

Source Symbol	Probability
a1	0.50
a2	0.25
a3	0.25

Using arithmetic coding, encode the message a1 a2 a3 a1. Show the resulting subinterval after processing each symbol of the message.

4. (a) Illustrate the functional block diagram of a general Image Compression system. 3
(b) Consider the following 3×3, 8-bit image of a vertical edge: 5

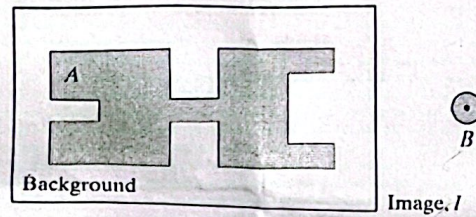
50 50 180

50 50 180

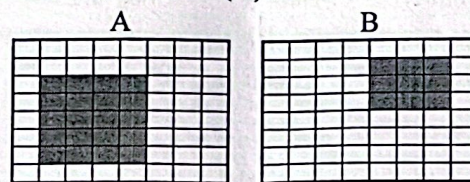
50 50 180

Encode the pixel sequence using LZW coding, obtained by scanning the image row by row and show all dictionary entries added. Assume the initial dictionary contains all individual pixel values as single-symbol entries.

5. (a) Define Dilation and Erosion in image processing. 2
 (b) Consider the following image I with the structuring element B . Show the final effect after performing opening and closing operations on I using B . 6



6. (a) Illustrate the following logic operations on the given binary images A and B: (i) NOT, (ii) AND, (iii) OR, and (iv) XOR 4



- (b) How does the size and shape of the structuring element (SE) affect the amount and manner in which an object shape changes in morphological operations? Explain with an example. 4
7. (a) State the key conditions that must be satisfied when partitioning an image's spatial region R into subregions $R_1, R_2, \dots, R_n$ during image segmentation. 2
 (b) State the convolution kernels used in the Sobel operator for edge detection. 2
 (c) Describe and illustrate the steps for region splitting and merging for image segmentation. 4